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Secondary Succession in Post-fire Scrub Dominated by *Acacia verticillata* (L'Hérit.) Willd. at Wilsons Promontory, Victoria

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Abstract

Senescence and death of 30-year-old post-fire *Acacia verticillata* scrub in areas originally supporting sclerophyll eucalypt forest were studied over a 6-year period at Wilsons Promontory, Victoria. The potential for eucalypt regeneration was also examined. On drier sites, *A. verticillata* has been replaced by drought-resistant *Kunzea ambigua* while on wet sites it has been replaced by waterlogging-tolerant *Melaleuca ericifolia*. In mid slopes *A. verticillata* is regenerating from soil seed in gaps in the field layer of *Pteridium esculentum* and *Goodenia ovata* and forming a patchy, uneven-aged stand. The composition of the mid-slope community is determined by the relative resistance of species to browsing and grazing. The failure of eucalypts to reclaim the scrub areas at this critical time has been due to poor seed crops, seed harvesting by ants and intense selective browsing. Exclosure plots have demonstrated the importance of herbivory in modifying this vegetation and indicate potential means of reafforestation without the intervention of fire.

Introduction

In the moist areas of Victoria, large areas of high forest have been converted to scrub of varying height, density and species composition by the fire regimes of the last century (Ashton 1981). Such dramatic changes have been ascribed to the fire sensitivity of the fast growing eucalypt overstorey and its relatively long non-flowering period. Thus, repeated fires at short intervals, even those of low intensity, will kill regeneration before seed has been produced. Alternatively, a similar result will be achieved if severe fires occur at a time of negligible seed storage in the crowns, or if severe drought or intense selective browsing removes or severely damages eucalypt regeneration.

The total area of potential sclerophyll forest on Wilsons Promontory (Fig. 1) is estimated to be 95 km^2 and includes a wide range of structure from dry to wet sclerophyll forest (Ashton 1981; Gill 1981). Much of the present forest is intermediate between these extremes and conforms to the 'damp sclerophyll' type as described by Opie *et al.* (1984). At Wilsons Promontory (Fig. 1) approximately 14% of the tall forests of *Eucalyptus obliqua* L'Hérit., *E. muelleriana* and *E. regnans** have been destroyed in the last 60 years by numerous fires.

Only 2·4% of the forested area currently consists of the tall open forest formation with tall and dense understoreys of shrubs, ferns and tree ferns. Of the forested areas converted to scrub, about 68% (9·0 km^2) are dominated by *Acacia verticillata* in damp sclerophyll areas and 32% (4·3 km^2) is dominated by *Pomaderris aspera, Bedfordia arborescens, Olearia argophylla* and *O. phlogopappa* in the wetter areas. Although

*Nomenclature and authorities follow Forbes *et al.* (1984).

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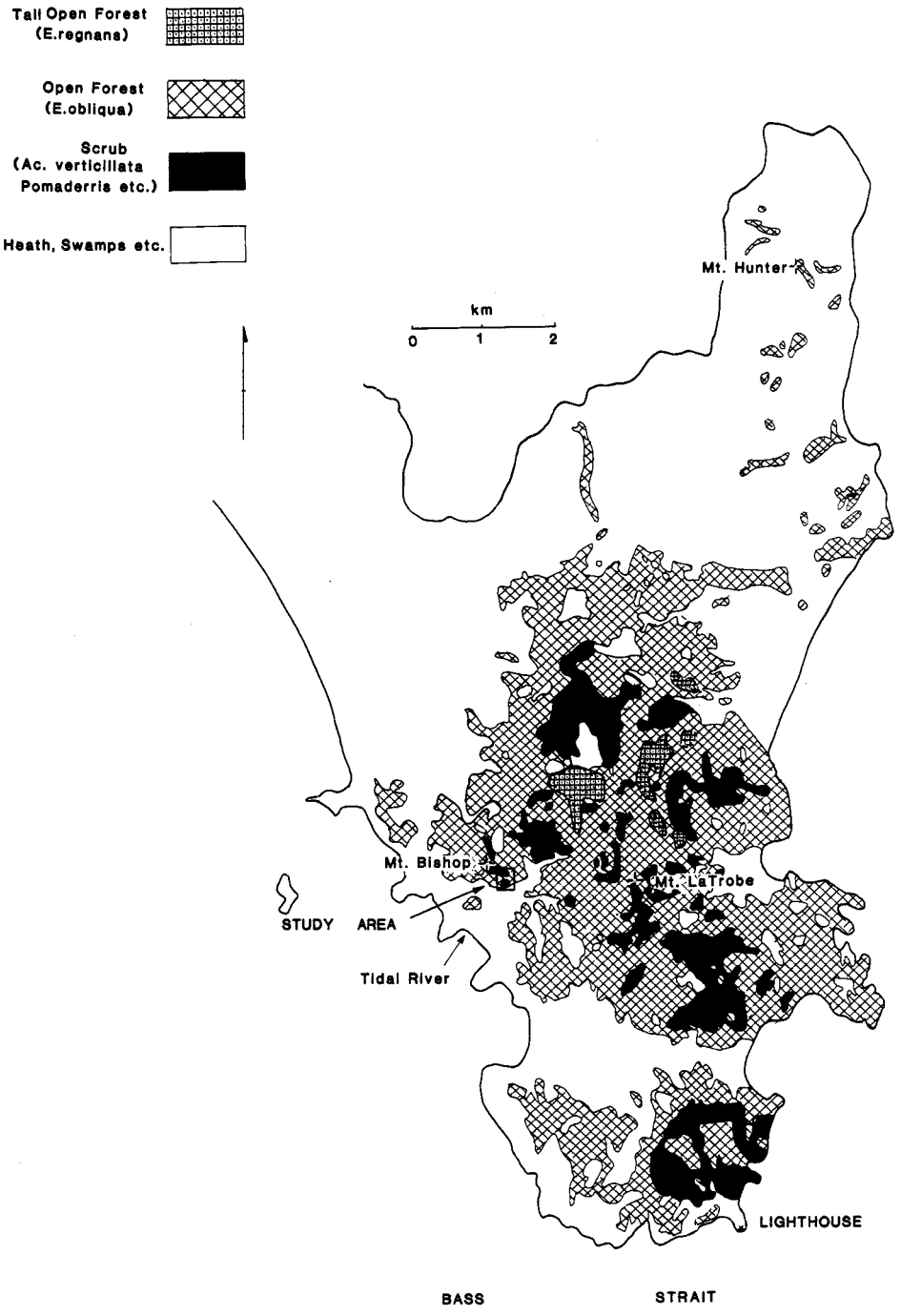


Fig. 1. The distribution of open forest (10–40 m) and tall open forest (>40 m) together with fire-induced scrub at Wilsons Promontory. For simplicity, scrub of wet sclerophyll species (*Pomaderris, Bedfordia*) has not been distinguished from damp sclerophyll (*Acacia verticillata, Coprosma*).

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E. obliqua is usually regarded as relatively fire resistant by virtue of its thick bark, it may be rendered susceptible if bark is depleted by repeated burning. Widespread introgression of this species with the thinner-barked *E. regnans* has resulted in trees with deciduous bark on the trunk and main branches; hence in many of the wetter sites, *E. obliqua* behaves as a fire-sensitive species (Smith 1970; Ashton 1981). Recent fires in the area (1927, 1939 and 1951) have produced a mosaic pattern of regeneration of the eucalypts and could have led to the development of scrub where the fires overlapped within the early non-flowering stage of the eucalypts, or where eucalypt death occurred at a time of meagre storage of capsules in the canopy, such as has occurred at Tidal River at times during the decade 1975–85. Extremely dense regeneration of *A. verticillata* in Lilly Pilly Gully followed the fires of 1951. Four years later this scrub was 2–3 m tall and virtually impenetrable. The ground stratum was almost devoid of



Fig. 2. The study site spur (L.H.S.) along Lilly Pilly valley looking towards Mt LaTrobe. (*a*) Dense *Acacia verticillata* in flower October 1967; (*b*) Same site 1986 when the scrub community had degenerated into a hummocky canopy of *Melaleuca, Banksia* and *Kunzea;* (*c*) Fenced and unfenced plots at commencement of experiment in 1980.

cover apart from mosses, sparse *Viola hederacea*, occasional clumps *Goodenia ovata* and rank *Tetrarrhena juncea*. After 10 years *A. verticillata* was mature and 4–5 m tall (Fig. 2*a*), but after 20 years it had begun to senesce (Fig. 2*b*). Death of stands commenced on the lower slopes and warmer aspects and by 1979–80 most stands were dead (Fig. 2*c*). The stems of the degenerating acacias were infested with numerous wood boring beetles, including *Notoceresium elongata* McKeown, *Pentarthrium australe* McKeown and *Menephilus coerulescens* Haag. It is possible that these and other insects have been responsible for the hastened decline of this even-aged generation. In 1978, regeneration from soil-stored seed of *A. verticillata* and *G. ovata* was occurring in open areas between patches of *G. ovata, Pteridium esculentum* and *T. juncea*. By 1979, browsing and grazing of vegetation was severe in certain areas due to an abundance of

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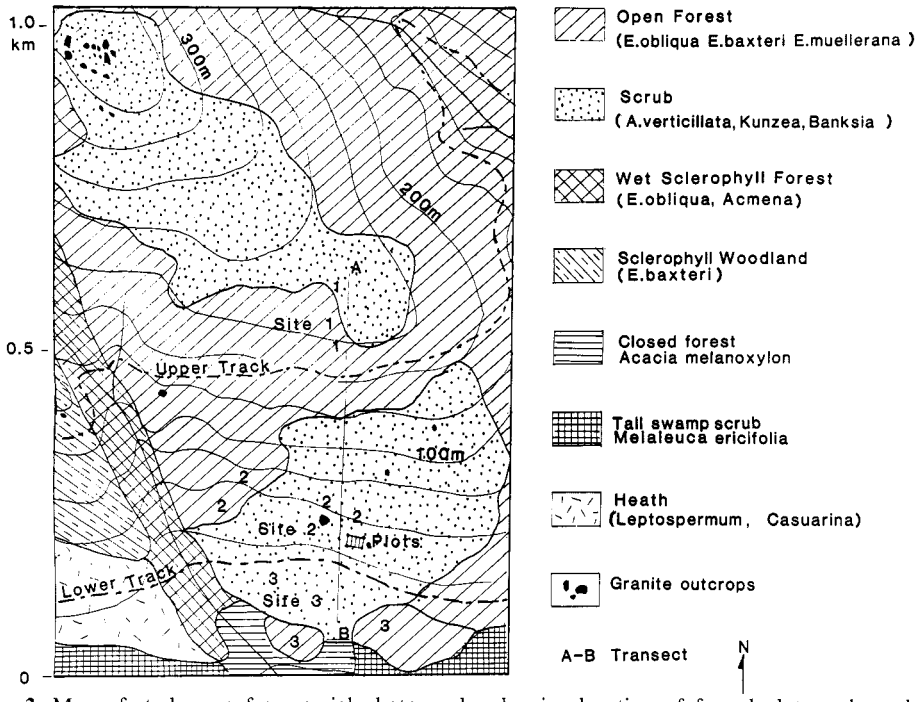


Fig. 3. Map of study area from aerial photography showing location of fenced plots and quadrat samples.

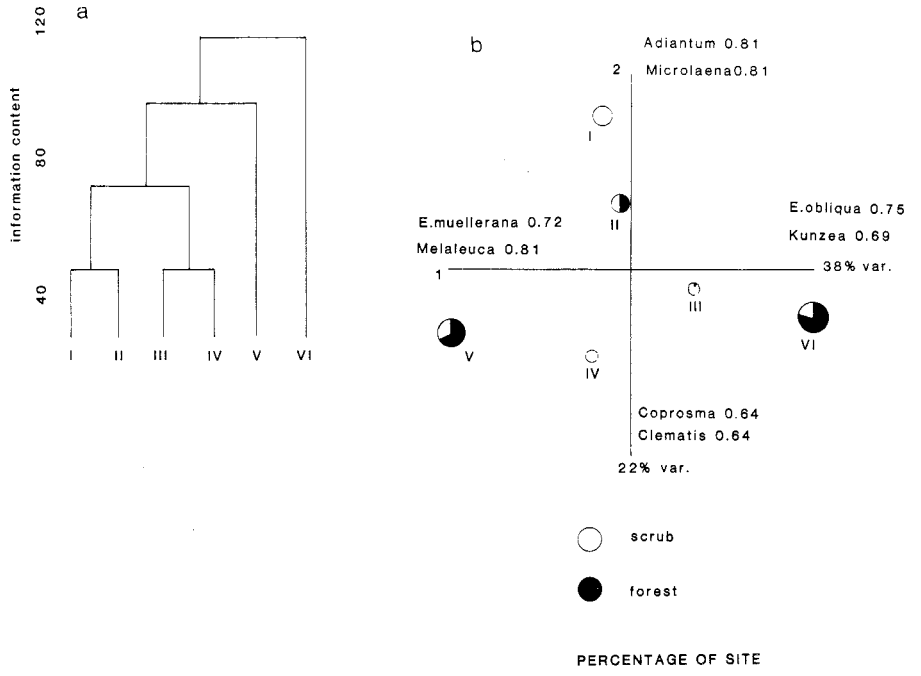


Fig. 4. (*a*) Dendrogram of classification into six groups (I–VI); (*b*) Ordination of groups (I–VI) indicating the proportion of forest and scrub quadrats in each. The two species with the highest correlation coefficients for axes 1 and 2 are indicated. The dimensions of the third axis are indicated by the size of circle.

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herbivores such as swamp wallabies (*Wallabia bicolor* Desmarest), wombats (*Vombatus ursinus* Shaw) and brushtail possums (*Trichosurus vulpecula* Kerr).

On the upper slopes, shrubs of *Kunzea ambigua* had invaded much of the degenerate scrub, whilst on the lower slopes the dominant, root-suckering species of the swamps, *Melaleuca ericifolia*, was invading much of the scrub area. The area chosen for detailed study in 1980, showed both these types of replacement. Eucalypt establishment without the intervention of fire in these communities is extremely rare.

A detailed assessment of the regeneration of the scrub began in 1979 (Chappill 1980) with the aim of predicting the future development of vegetation in these areas. The study site was near Tidal River, adjacent to the Lilly Pilly Gully walking track. The site location and the distribution of vegetation types as determined from aerial photography is indicated in Fig. 3. The study included description of the structure and species composition of the scrub, a field trial examining the possible survival of eucalypt seedlings and a survey of other species regenerating at that time.

Materials and Methods

Vegetation Structure and Composition

Floristic composition was assessed in 1983 by gathering data from 78 quadrats (10 by 2 m) along three transects from the edge of the Tidal River swamp to the *K. ambigua*-dominated scrub on the upper slopes of Mt Bishop. The pattern in these data was revealed with the aid of the computer-based hierarchical agglomerative classification program MACINF with presence and absence data (Lance and Williams 1967) and principal coordinates analysis (PCOA) of the data matrix produced by MACINF. The programs were accessed in the CSIRO TAXON package (Ross 1983).

Profiles were drawn in three representative areas to display the structure of the vegetation.

The relative areas occupied by light (<25%), medium (25–75%) and heavy (75–100%) cover classes of the field layer were assessed from intercepts on 16 lines 10 m long on the lower slopes of the degenerate scrub in 1983.

Regeneration of Eucalypts in the Scrub

An experimental plot 200 m^2 was chosen in a homogeneous area of typical degenerate scrub on the lower slopes (Fig. 3). It was divided into four 10 by 5 m subplots, two of which were fenced with 4 cm wire-mesh netting to a height of 1 m. The fences excluded wallabies and wombats but not possums. Seventy-two seedlings of each of the three eucalypt species which grow nearby, i. e. *E. obliqua* and *E. muelleriana** in the moister areas and *E. baxteri* in the drier areas, were planted in December 1979 in a random block design in three cover classes of vegetation: natural vegetation, vegetation clipped to 15 cm height in a radius of 15 cm around the seedlings and areas completely cleared of vegetation to a radius of 15 cm around the seedlings. At each spot the seedlings were assigned randomly at three points of a triangle, 10 cm apart. This resulted in 12 replications of each treatment and species combination. Seedlings were first measured on 31/12/1979 and the changes in height, growth and survival were recorded at approximately monthly intervals until mid September 1980, and then at approximately yearly intervals until 1986.

Additional plantings were made of *E. obliqua* seedlings 1–1·5 m tall in the dense vegetation surrounding the experimental plots to determine whether larger plants would be better able to establish in the scrub. Seeds of the three eucalypts were sown separately in 10 small microsites of disturbed soil 30 cm in diameter outside the plots, to evaluate the level of predation by insects or mammals on very young stages and the success of germination. All seed was lightly covered by soil, as it had been shown by A. N. Andersen (personal communication) in nearby sites that insect predation by, for example, the ants *Rhytidoponera victoreae* Andre and *Chelaner sculpturatus* Clark. could account for losses of 33–83% of the seed placed on the soil surface within 2 weeks.

Regeneration of A. verticillata *and Other Species in the Scrub*

A number of *A. verticillata* seedlings in the fenced and unfenced plots of the eucalypt field trial was

*Seed of *E. obliqua* and *E. muelleriana* was obtained from similar environments at Neerim and Yarram respectively, owing to a lack of seed at Wilsons Promontory in 1979.

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Table 1. The species composition of computer-based groups. (Groups are arranged in topographic order)

(*a*) Percentage frequency of groups (I–VI) in forest and scrub on a topographic gradient. The number of quadrats in each group is indicated in parentheses

	VI(19)	III(14)	I(22)	II(12)	V(18)	IV(5)
Upper slope						
Forest	58	7	—	—	—	—
Scrub	21	58	—	—	—	—
Mid slope						
Forest	21	—	—	42	6	—
Scrub	—	21	100	42	22	—
Lower slope						
Forest	—	—	—	8	61	—
Scrub	—	14	—	8	11	10
Total	100	100	100	100	100	10

(*b*) Percentage frequency of species in each group (I–VI)

	VI	III	I	II	V	IV
Trees						
E. obliqua	79	—	—	17	5	—
E. muelleriana	—	—	—	17	61	—
E. baxteri	5	7	—	8	—	—
Shrubs–small trees						
Melaleuca ericifolia	—	36	60	58	61	100
Kunzea ambigua	79	57	70	83	5	—
Acacia verticillata	10	57	40	58	33	—
Banksia marginata	5	57	70	50	5	20
Goodenia ovata	84	50	100	92	94	80
Olearia lirata	—	—	—	17	—	—
Leptospermum juniperinum	16	—	—	8	—	—
Pultenaea daphnoides	—	7	10	—	—	—
Coprosma quadrifida	—	7	—	17	—	—
Lianes						
Clematis aristata	5	21	—	—	100	72
Pandorea pandorana	5	29	—	58	—	20
Comesperma volubile	5	14	—	—	5	—
Ferns						
Pteridium esculentum	84	79	67	90	100	—
Adiantum aethiopicum	—	—	8	100	5	—
Cyathea australis	—	—	8	—	5	—
Graminoids						
Tetrarrhena juncea	100	86	67	70	83	100
Microlaena stipoides	—	—	8	70	22	20
Lepidosperma longitudinale	84	71	25	80	94	40
Gahnia clarkei	—	—	—	—	11	—
Forbs (dicotyledons)						
Viola hederacea	37	64	100	90	83	60
Gonocarpus teucrioides	37	—	58	20	44	—
Oxalis corniculatus	—	—	50	40	22	60
Hydrocotyle hirta	—	—	8	—	11	80
Lobelia anceps	—	—	33	—	—	20
Gnaphalium japonicum	5	—	8	—	—	—
Stellaria flaccida	—	—	—	—	61	—
Bryophytes and lichens						
Bryum billardieri	42	64	42	10	11	80
Thuidium laeviusculum	26	64	8	—	—	80
Parmelia conspersa	—	7	17	20	5	—
Lophocolea ulothrix	—	—	—	—	11	—
Sematophyllum sp.	—	7	—	—	—	—
No. of species	18	19	27	16	24	14

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regularly measured. As the seedlings were not of a uniform age at the first measurement, the percentage increase in height from the first measurement rather than mean heights was calculated.

The vegetation cover and density in the fenced and unfenced plots was assessed by vertical point quadrats using pins 1·5 mm diameter and 2 m tall. These analyses were repeated at 1–2 year intervals from 1980 until 1986. The density of the major shrub species was also assessed outside the plots by counting numbers of individuals in 16 10 by 2 m quadrats in the lower mid-slope area in 1982 and 1983.

In early January 1980, five plots in the scrub, each 1 m^2, were cleared of vegetation and litter with minimal disturbance to the topsoil. In June, July and September of the same year the identifiable seedlings in a central square, 30 by 30 cm, were counted and removed. In early February, soil was collected from 10 random plots in the same community for examination of seed germination under more favourable conditions of moisture and protection in seed trays placed in the open on a rooftop at the University of Melbourne. Three depths of soil were sampled; 0–1 cm, 1–2 cm and 2–3 cm. The soil was sieved and spread to a depth of 1 cm over sterile coarse sand in seed trays 28 by 32 cm. Germination was monitored between March and July by counting seedlings and removing them as they were identified.

In 1983, the relative frequency of herbivory on various species was assessed in 16 10 by 2 m quadrats on the lower slopes of the degenerate scrub, each species being classified according to the relative intensity of damage sustained (i. e. no damage, light, heavy). In the course of floristic assessment from the ridge top to the swamp, faecal pellets within quadrats were counted and classified according to species of origin.

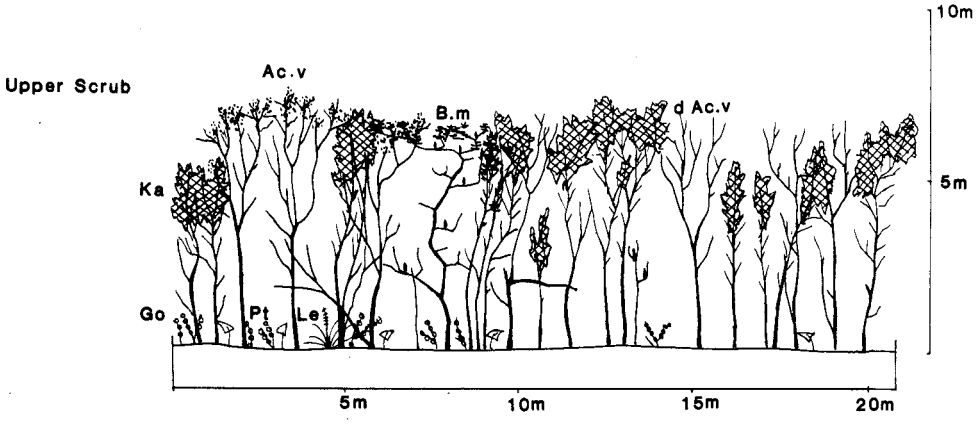


Fig. 5. Profile of upper slope scrub of *Kunzea* and *Banksia*. Ac.v, *A. verticillata*; dAc.v, dead *A. verticillata*; B.m, *Banksia marginata*; Ka, *Kunzea ambigua*; Le, *Lepidosperma elatius*; Go, *Goodenia ovata*; Pt, *Pteridium esculentum*.

Results

Vegetation Structure and Composition

The classification of the floristic composition of quadrats (10 by 2 m) produced six groups (Fig. 4*a*) which could be interpreted in relation to topography. Although some groups (I, V) comprised only quadrats from scrub vegetation (Table 1), others (II) were equally distributed between scrub and forest or indicated a distinct preference for one or the other (VI, III). The greatest number of species occurred in forested lower-mid slopes (I), the lowest in wet scrub (IV). Most groups were characterised by gradations of species frequencies although the predominant community in the mid-lower slopes of *Melaleuca-Kunzea-Banksia-Acacia* was distinguished by the high frequency of *Adiantum aethiopicum* in the *Pteridium-Goodenia* field layer. The first two axes of the ordination of the groups (Fig. 4*b*) accounts for 60% of the quadrat variation. Axis 1 appears to be related loosely to topographic factors such as soil moisture gradients.

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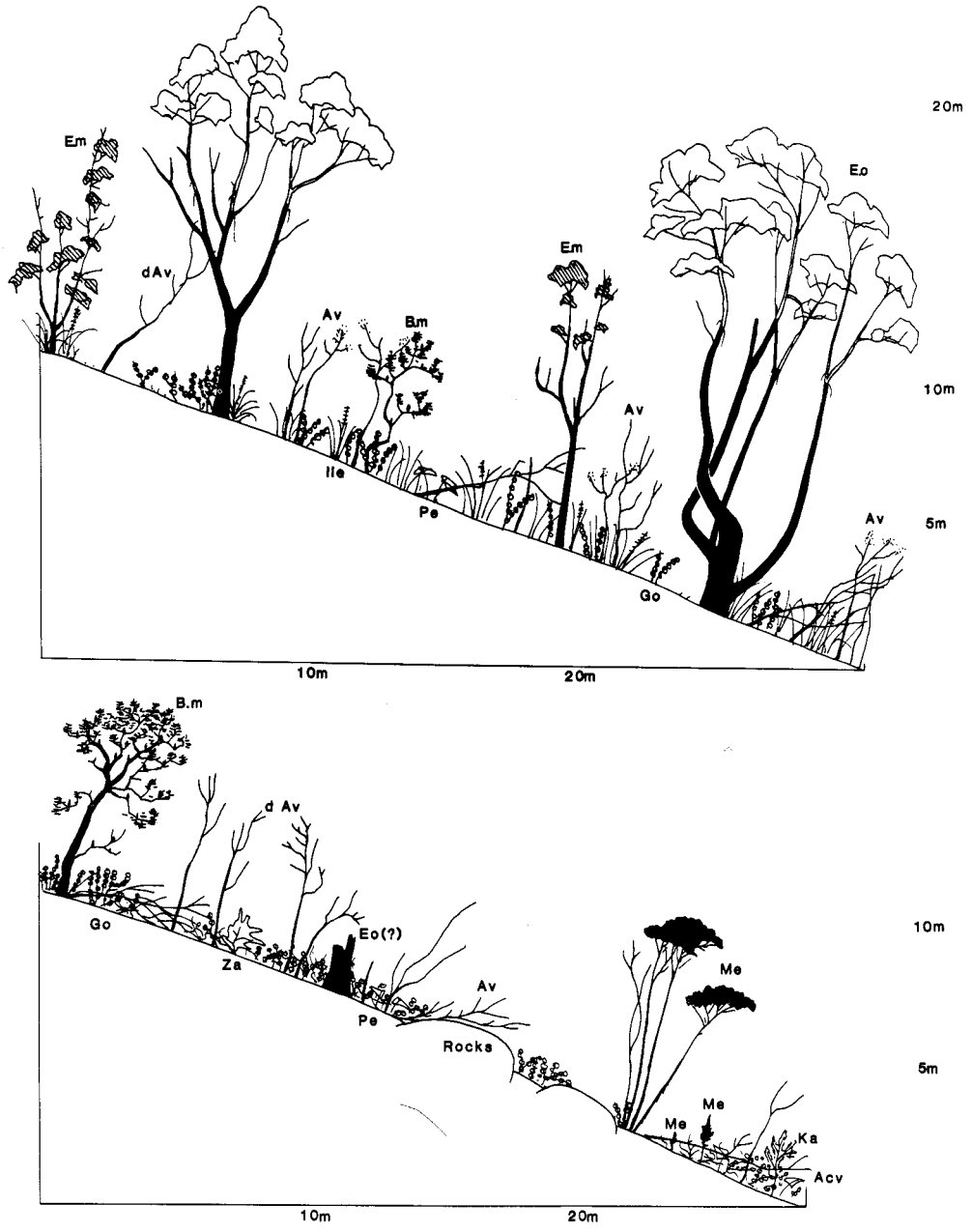


Fig. 6. (upper) Profile of open forest on mid slope. (lower) Profile of degenerate scrub on mid slope. Av, *A. verticillata*; d Av, dead *A. verticillata*; Eo, *E. obliqua*; Em, *E. muelleriana*; Me, *Melaleuca ericifolia*; Za, *Zieria arborescens*. Other symbols as in Fig. 5.

The structure of the scrub and forest communities along the topographic gradient is shown in the profiles (Figs 5, 6 and 7). The cover of the field layer of *Pteridium, Goodenia* and *Tetrarrhena* varied greatly, from very low under dense shrubs to very high in open areas in the moister sites. On the lower slopes, in open scrub where *A. verticillata* had died out, 42% of the area had cover values of >75% (mean height

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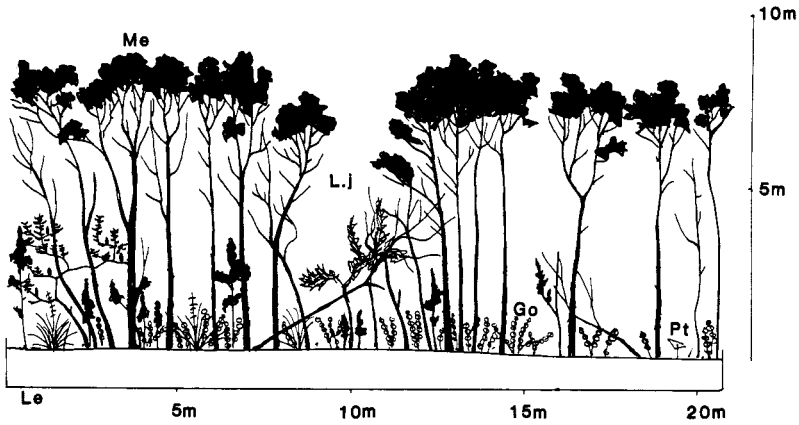


Fig. 7. Profile of *Melaleuca* scrub on lowermost slope. Lj, *Leptospermum juniperinum*. Other symbols as in Figs 5 and 6.

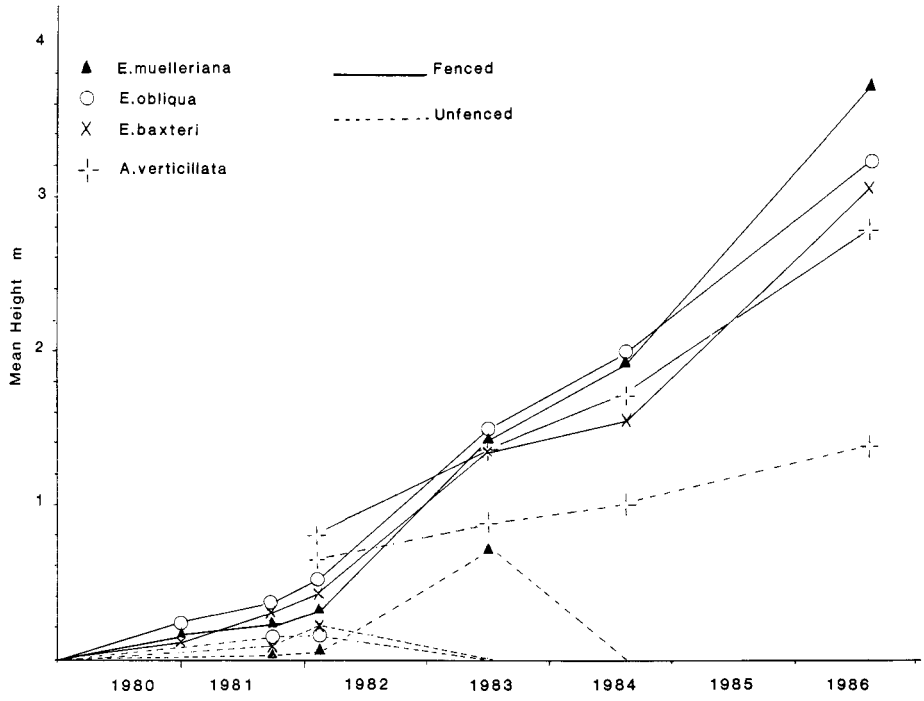


Fig. 8. Mean height growth of eucalypt species and *Acacia verticillata* in fenced and unfenced plots 1980–86.

125 cm) whereas only 27% had cover values of <25% (mean height 43 cm). Seedling regeneration was confined to the more open areas, although there it was prone to greater browsing pressure. Since the field layer has continued to thicken, the area available for regeneration is likely to diminish.

Regeneration of Eucalypts in the Scrub

To allow statistical comparisons of height growth to be made between the transplanted seedlings of the three species in the fencing treatments, it was necessary to pool

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the data of all microhabitats because of the low survival in the lighter cover classes. Two-way analyses of variance were performed after logarithmic transformation of each set of measurements. No significant differences between species were observed. However, in 1986 the tallest of the dominant-crowned individuals of *E. muelleriana* (400 cm) far exceeded the tallest *E. obliqua* (256 cm) and *E. baxteri* (285 cm). A longer-term study may show that *E. muelleriana* will come to dominate the site in future years.

The effect of the fencing treatment was significant by April 1980, when all species showed greater height growth (at $P = 0{\cdot}05$) in the fenced than unfenced plots. Height growth of the eucalypt species in the fenced plots increased markedly in the second and third years, in spite of the severe drought of 1982 (Fig. 8).

The difference in survival between different species of eucalypt during the 6 years of the study were marked. *E. muelleriana* showed consistently higher survival rates than the other species after the first year and *E. obliqua* had the fewest seedlings surviving. After 12 months, only about 20% of the unfenced plants had survived, and after 2 years one plant of *E. muelleriana* remained, and that specimen remained only for an

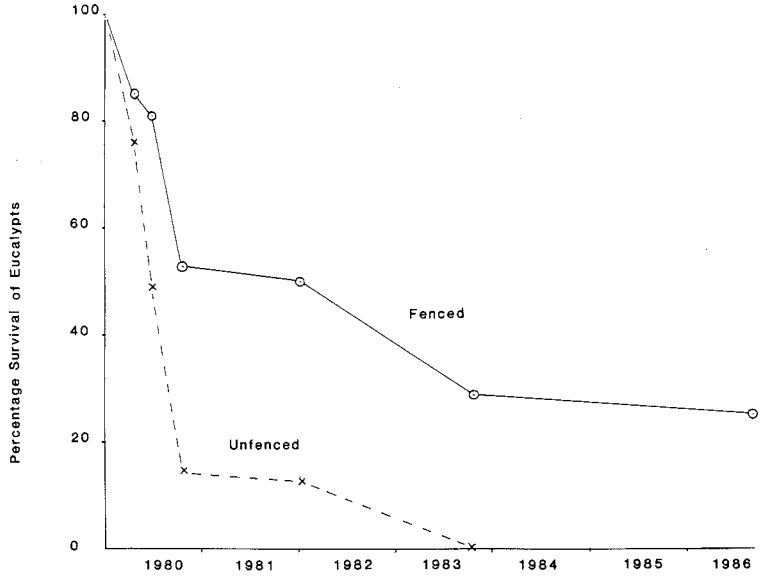


Fig. 9. Percentage survival of all eucalypts in fenced and unfenced plots 1980–86.

additional few months. The survival of plants in the fenced plots declined steadily to about 20% over the ensuing 3 years. The pooled data for all species in fenced and unfenced plots are shown in Fig. 9. *E. obliqua* was more severely attacked by groups of sawfly larvae than the other species, and *E. muelleriana* maintained large and horizontally displayed juvenile foliage for at least the second growing season.

The fate of large eucalypt seedlings planted at random in thick scrub and ground stratum in the mid-slope area indicated that plants 1–2 m tall could be located, pulled down and severely browsed, most probably by wallabies. After 3 years, all plants had been destroyed, even those in thick regrowth of *M. ericifolia* and *G. ovata*. The last surviving plant, 2 m tall in 1985, had disappeared without trace 1 year later.

Seeds of the three eucalypt species sown outside fences in mid to late autumn, germinated within 2 weeks and reached the 2-leaf stage and heights of 2–8 cm by the following spring. By December, browsing was evident in four out of 21 plants. By the following autumn all plants had disappeared.

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Regeneration of A. verticillata *and Other Species in the Scrub*

The development of the ground stratum in response to the exclusion of the larger herbivores from the fenced areas was dramatic (Table 2). The most conspicuous effect in the first 2 years was the increase in cover and height of the scrambling grass, *T. juncea* and the raspwort, *Gonocarpus teucrioides*. Both of these species declined sharply during the drought year of 1982 and only the former recovered its prior vigour (Fig. 10). *P. esculentum* behaved similarly to *T. juncea*, but *G. ovata* declined steadily as its associates increased. The cover repetition of species sensitive to browsing tended to be higher in the fenced plots, although *Coprosma quadrifida* did not conform, due to its development of compact, intricate spiny branches under heavy browsing (Table 2). Species diversity was greater in the fenced plots (26 species) than the unfenced plots (20 species).

In 1986 the increasing cover of eucalypts in the fenced plots appears to have reduced the vigour of the field layer. In the unfenced plots *T. juncea* decreased markedly, probably as a result of wombat grazing. The field layer, therefore, became increasingly dominated by non-grazed *P. esculentum* and *G. ovata* and with scattered spiny bushes of *C. quadrifida* and *A. verticillata*. *K. ambigua* and *M. ericifolia* which are increasing in density in these areas are scarcely browsed, although occasionally killed by wood-boring insects on the lower stems.

The density of natural regeneration of the major woody species in the mid-slope area of the scrub in 1982 and 1983 is shown in Table 3. *M. ericifolia* is regenerating vigorously from root suckers and *A. verticillata* is also regenerating well from soil-stored seed. The later species flowered and set seed abundantly in 1984 and thus further replenishment of the soil seed bank is likely to have occurred in 1985. The reduction of height and density of *A. verticillata* in the presence of browsing was evident in 1986.

The observed germination of soil-stored seed in the field and in seed trays is given in Table 4. The number of seedlings germinating from the seed trays was greatest from the upper 1 cm sample, but the species composition was similar for each depth; therefore, the results were pooled and adjusted to show numbers of germinated seedlings per square metre of soil to a depth of 3 cm. Most species showed greater germination in the trays than in the field, with the exception of *Hydrocotyle hirta* and *V. hederacea*. Some species such as *Bulbine bulbosa, Calystegia marginata* and *Schoenus apogon* appeared in the trays but were not observed in the natural vegetation. The notable absence of *T. juncea*, may be due to heavy grazing preventing normal flowering, since this species is normally present as soil seed in wet forests in central Victoria (Ashwell 1981).

The species subject to grazing and browsing are indicated in Table 5. Scat analysis indicates that black-tailed wallabies and the common wombat are the major herbivores and differ in their relative abundance along the topographic gradient. This will affect the proportion of grazing and browsing in different sites. Brushtail possums were present, and may have been responsible for minor browsing of eucalypt seedlings in 1980 within the fenced plots. Hog deer (*Axis porcinus*) are present in the area but do not appear to be as important as the marsupials at this time.

It is clear that the species which are most successful in the regeneration in this area have means of resistance to grazing pressure. Thus, both *C. quadrifida* and *A. verticillata* are protected to some extent by spines, whilst *G. ovata, K. ambigua* and *M. ericifolia* appear to be unpalatable to the mammals in the area, although insect attack may occur. In many places it is evident that the greatest amount of damage occurred in the vicinity of animal runs and in more open spaces between the denser vegetation. Dicotyledonous herbs, such as *V. hederacea, Stellaria flaccida* and *G. teucrioides* were severely grazed, but *P. esculentum* and *G. ovata* were conspicuously undamaged, although minor insect attack was evident.

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Table 2. Percentage cover of vegetation in fenced and unfenced plots, Tidal River 1979–86, by means of vertical point quadrats (1·5 mm diameter)

Cover repetition >1·0 in parentheses. No. of points 110–240. *Seedlings planted 2 cm high 1979

	Fenced					Unfenced				
	1979	1982	1983	1984	1986	1979	1982	1983	1984	1986
Shrubs, trees species										
*Eucalyptus obliqua**			4·5(1·4)	5·7(3·6)	3·0(3·0)					
*E. muelleriana**			4·5(2·8)	3·4(2·4)	5·0(4·1)					
*E. baxteri**		0·9	1·8(2·0)	1·5(2·0)	8·0(5·5)					
Acacia verticillata		8·6(3·1)	14·5(2·8)	25·5(3·7)	35·0(3·8)	1·0		8·8(1·9)	15·4(2·4)	27·0(2·5)
Kunzea ambigua			2·0(1·5)	1·1(1·7)	4·0(1·8)		7·0(3·0)	9·2(1·8)	5·4(1·8)	7·0(1·4)
Melaleuca ericifolia			1·4(1·7)	0·8(2·0)	1·0	2·0(3·0)	4·5(3·0)	12·1(3·6)	9·6(2·6)	14·0(3·6)
Coprosma quadrifida		0·9	3·5(2·6)	3·8(2·0)	3·0(2·7)		0·9(3·0)		1·3(4·3)	6·0(3·3)
Goodenia ovata	53·0(1·7)	57·7(1·8)	30·4(2·1)	32·3(2·7)	38·0(2·0)	37·0(1·9)	60·2(1·8)	46·8(2·2)	40·8(1·9)	37·0(1·9)
Helichrysum dendroideum				1·5(2·5)	1·0					1·0
Pultenaea daphnoides				0·8						
Bursaria spinosa	2·0									
Banksia marginata					2·0		2·9(3·3)			1·0
Ferns										
Pteridium esculentum	16·0(2·6)	36·6(2·5)	37·5(2·2)	59·3(2·6)	53·0(2·6)	20·0(2·1)	42·7(3·2)	24·3(2·1)	55·8(2·1)	41·0(2·1)
Adiantum aethiopicum					1·0					
Culcita dubia	1·0									
Graminoids										
Tetrarrhena juncea	56·0(1·4)	78·8(2·5)	55·5(2·2)	70·3(2·7)	54·0(2·6)	37·0(1·3)	29·1(1·3)	18·0(1·5)	12·7	22·0(1·3)
Lepidosperma longitudinale	1·0	1·9	10·2(1·8)	8·3(2·8)	20·0(2·8)	1·0	1·9	8·3(2·0)	0·4	1·0
L. elatius			7·4(2·0)							
Gahnia sieberiana		0·9	1·1(1·1)	6·0(3·2)	2·0(1·1)					1·0
Poa tenera							2·9(1·7)			
Danthonia penicillata										1·0
Microlaena stipoides	0·5	0·5							7·1(1·3)	

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Lianes										
Pandorea pandorana	1·0	5·8	8·1(1·3)	16·0	9·0(1·2)	5·0	0·9	0·1	2·9(1·1)	5·0
Comesperma volubile		12·5(1·7)	2·5(2·0)	12·1(1·8)		2·0	3·9			
Billardiera scandens		4·8(1·4)	4·9(1·9)	6·8(1·8)	8·0(2·6)					
Cassytha pubescens			12·0(1·7)	13·7(2·8)						
Forbs and dicotyledons										
Gonocarpus teucrioides	3·0	54·8(1·9)	17·0(1·9)	2·3(1·7)		3·0	9·7(1·9)	3·3(1·2)	0·8	
Viola hederacea	6·0	1·9	1·7	5·3(1·1)	3·0(1·3)	3·0	11·6(1·1)	5·8	7·1	3·0
Stellaria flaccida									0·4	
Hydrocotyle hirta	1·0					2·0				
Sigesbeckia orientale	1·0									
Oxalis corniculatus	1·0								0·4	
Dichondra repens		0·9								
Mosses			1·0	3·0	2·0		1·9	10·0	7·5	7·0
Litter only	15·0	4·0	5·6	4·2	1·0	24·0	4·8	12·5		8·0
Bare ground only	1·0				1·0	2·0	1·9	1·2	8·0	10·0
Rock								0·1		2·0
Total vegetation	86·0	96·0	94·4	95·8	98·0	76·0	92·0	86·2	85·8	90·0
Number of species	13	15	19	20	18	11	13	10	14	14

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The effects of herbivore exclusion are complex, due to relative growth and competition of existing species. Thus, the demise of *G. teucrioides* in 1985–86 in fenced plots may have been due to competition. Diversity was initially higher in the fenced plots in the early stages of exclusion, but the number of species declined as the density of the more vigorous species increased.

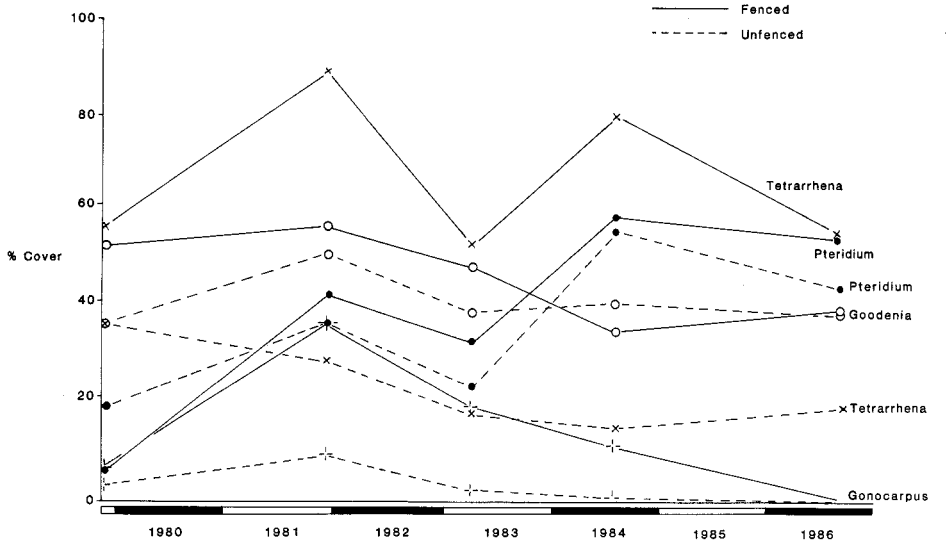


Fig. 10. Cover values of major understorey species in fenced and unfenced plots 1980–86.

Table 3. Density of major shrubs (No./ha) (from 16 quadrats, each 10 by 2 m) on lower to mid slope, degenerate *Acacia* scrub, 1982 and 1983

*Stump remaining from 1951 fire

Species	Original stand		Regeneration	
	Dead	Living, mature	Seedling	Suckers
Eucalyptus sp. (*obliqua*?)	31*	—	—	—
Acacia verticillata	2968	—	313	—
Melaleuca ericifolia	—	250	0	2562
Coprosma quadrifida	—	—	108	—
Kunzea ambigua	—	31	156	—
Banksia marginata	—	31	0	31
Leptospermum juniperinum	—	—	31	—
Zieria arborescens	—	31	—	31
Hakea sericea	—	31	—	—
Total	2997	374	608	2624

Discussion

Purdie and Slatyer (1976) have pointed out that a single model of secondary succession is unlikely to describe adequately a given sere, and this is true for the post-fire scrubs at Wilsons Promontory. The development of sclerophyll open forests is the product of site quality and fire regime (Gill 1975). In the scrub areas the normal post-fire succession has been altered by the removal of the *Eucalyptus* overstorey component in the last 100 years by a regime of frequent, severe fires associated with European settlement (King 1963).

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Table 4. Density of soil seed germination from scrub (*Acacia–Kunzea–Melaleuca*) in lower slope sites compared to frequency of species in the vegetation in 15 plots 10 by 1 m

Species	Frequency in vegetation (%)	Field germination/m^{2A} 3 June	6 July	12 Sept.	Seed tray germination/m^{2B} 20 Mar.	17 Apr.	29 July
Shrubs							
Goodenia ovata	100	77·7	124·3	128·3	101·1	267·5	321·9
Banksia marginata	67	—	—	—	—	—	—
Coprosma quadrifida	57	—	—	—	—	—	—
Acacia verticillata	50	7·4	8·9	8·9	18·7	23·3	23·3
Leptospermum juniperinum	40	—	—	—	—	—	—
Kunzea ambigua	20	—	—	—	—	—	—
Hakea sericea	10	—	—	—	—	—	—
Pultenaea daphnoides	7	—	—	—	—	—	—
Climbers							
Pandorea pandorana	53	—	—	—	—	1·1	1·1
Billardiera scandens	47	—	—	—	—	—	—
Comesperma volubile	43	—	—	—	—	—	—
Clematis aristata	40	—	—	—	—	—	—
Calystegia marginata	—	—	—	—	1·1	1·1	1·1
Forbs							
Viola hederacea	100	310·8	435·1	515·0	—	—	164·3
Hydrocotyle hirta	100	113·2	202·0	233·1	31·1	154·3	178·7
Gonocarpus teucrioides	100	31·1	48·8	64·4	27·8	113·2	119·9
Galium gaudichaudii	100	—	—	—	—	3·3	4·4
Oxalis corniculatus	93	4·4	8·9	11·1	31·1	61·1	72·2
Wahlenbergia quadrifida	23	—	—	8·9	—	1·1	1·1
Poranthera microphylla	10	—	—	—	—	—	—
Bulbine bulbosa	—	24·4	40·0	40·0	30·0	93·2	96·6
Pelargonium australe	—	—	—	—	12·2	20·0	20·0
Solanum aviculare	—	—	—	—	1·1	9·8	9·8
S. nigrum	—	—	—	—	3·3	3·3	3·3
Graminoids							
Lepidosperma longitudinale	93	—	—	—	—	—	—
Tetrarrhena juncea	70	—	—	—	—	—	—
Microlaena stipoides	33	—	—	—	—	—	—
Poa tenera	33	—	—	—	—	—	—
Danthonia racemosa	14	—	—	—	—	—	—
Schoenus apogon	—	2·2	175·4	175·0	—	—	926·9
Gahnia sieberiana	—	—	—	—	—	—	4·4
Ferns							
Pteridium esculentum	93	—	—	—	—	—	—
Adiantum aethiopicum	60	—	—	—	—	—	—
Cyathea australis	33	—	—	—	—	—	—

[A]Germination in five bare soil plots (30 by 10 cm) January to September 1980.
[B]Germination from 10 trays (30 by 10 cm to depth 3 cm) February to July 1980.

The early stages of post-fire succession are strongly influenced by the composition of the previous vegetation or by its legacy of soil-stored seed. In this sense the succession is of the 'initial floristic composition' type of Egler (1954). Of the species present in the study area, 13% recover vegetatively from trunk buds, root stock or root suckers and 70% regenerate from seed, most of which is present in the soil seed bank. The relative density of seedlings and root suckers is shown in Table 3. The invasion of the senescent

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A. verticillata stage by other shrubs suggests that in the local extremes of the habitat range, the succession conforms to the 'tolerance' model of Connell and Slatyer (1977). Thus, during the maturation of the dominant *A. verticillata* generation the upper slopes were invaded by the drought-resistant shrub *K. ambigua*, possibly by wind dispersal from fire-protected rocky outcrops. On the lowermost slopes *M. ericifolia* had regenerated by means of both root suckers and seedlings and had developed a distinct under-

Table 5. Percentage frequency of occurrence of browsing intensity in 13–20 plots (10 by 2 m) in field layer of the lower slope region

Browsing intensity: light, a few leaves or shoots browsed; medium, many leaves or shoots browsed; heavy, most leaves or shoots browsed. i, insect; k, koala

Species	*Kunzea* scrub upper slope (n = 13)				*Acacia–Melaleuca* scrub lower slope (n = 20)			
	Unbrowsed	Browsed			Unbrowsed	Browsed		
		Light	Medium	Heavy		Light	Medium	Heavy
Woody								
Eucalyptus muelleriana	—	—	—	—	—	—	—	5k
Acacia verticillata	46	—	23	—	55	30	15	—
Kunzea ambigua	31	—	—	—	55	25	—	—
Coprosma quadrifida	31	—	—	—	—	—	—	20
Pultenaea daphnoides	77	—	—	—	—	—	—	—
Melaleuca ericifolia	—	—	—	—	30	40	20	—
Leptospermum juniperinum	—	—	—	—	70	—	—	10
Goodenia ovata	24	38i	15i	—	35	45i	—	—
Banksia marginata	23	—	—	—	5	—	—	—
Hakea sericea	—	—	—	—	5	—	—	—
Hedycarya angustifolia	—	—	—	—	—	5	—	—
Correa reflexa	7	8	—	—	—	—	—	—
Graminoids								
Tetrarrhena juncea	15	23	31	31	15	—	15	70
Gahnia sieberiana	44	—	15	—	55	10	10	10
Danthonia 'pilosa'	—	—	—	—	—	5	—	—
Lepidosperma elatius	31	—	—	—	—	—	—	—
Climbers								
Pandorea pandorana	—	—	—	—	5	—	—	—
Clematis aristata	15	8	—	—	5	5	5	—
Comesperma volubile	69	8	8	—	70	5	10	—
Ferns								
Pteridium esculentum	—	—	—	—	65	10i	20i	—
Adiantum aethiopicum	—	—	—	—	80	15	—	—
Forbs								
Gonocarpus teucrioides	69	23	—	—	70	15	—	15
Viola hederacea	54	15	—	—	5	5	—	10
Hydrocotyle hirta	—	—	—	—	15	—	—	5
Galium gaudichaudii	—	—	—	—	10	—	—	—
Bulbine bulbosa	—	—	—	—	5	—	—	—
Oxalis corniculatus	—	—	—	—	5	—	—	—

storey. Therefore, with the onset of senescence and death of *A. verticillata*, the vegetation of the drier sites was rapidly dominated by *K. ambigua* and that of the wetter sites by tall *M. ericifolia*.

In terms of dynamics, the mid and lower slopes represent a relatively stable stage since *A. verticillata* is regenerating; however, much will depend on the growth, toler-

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ance and longevity of other dominants such as *B. marginata, C. quadrifida, K. ambigua* and *M. ericifolia*. The successfully regenerating species are either unpalatable or spiny at shoot maturity. The importance of the soil seed bank and the ability of species to avoid the sensitive seedling stage by vegetative spread is clearly seen in the potentially dense field and ground strata of this vegetation. In this area the scattered individuals of *A. verticillata* flowered after 5 years, thus restoring to some extent the soil seed bank of this species.

The mid-slope areas show the greatest diversity both structurally and floristically. With the demise of the *A. verticillata* stratum the high percentage of bare ground was subject to the effects of insolation, and regeneration of soil seed commenced. The patchy regeneration of *A. verticillata*, which commenced about 10 years ago, is continuing, but the area available is being diminished as the field layer of *P. esculentum* and *G. ovata* increases in area, density and height. Therefore, at this stage, *A. verticillata* has contracted its range to the mid slopes where the previously dense, even-aged stand is being replaced by a patchy, uneven-aged community, with an age span of more than 15 years. Future cycles of regeneration may further exaggerate this tendency and produce a regeneration mosaic, sens. Watt (1947), although severe fires in the next century may again restore an even-aged community from the soil seed bank over the whole topographic range.

Only under very rare instances do eucalypt seedlings survive without the intervention of fire. The experimental plots demonstrate conclusively the overwhelming importance of the herbivore factor in preventing the regeneration of *Eucalyptus* species under otherwise favourable conditions. Much seed is harvested by insects, and seedlings beyond the two-leaf stage are readily browsed. Over the first few years little discrimination occurred between the three contending species in the area — *E. obliqua, E. muelleriana* and *E. baxteri*. When fenced, the latter two species have survived better in the luxuriant ground stratum, possibly as a consequence of their longer period of horizontally disposed juvenile foliage. However, these data may be biased by the different seed sources for *E. obliqua* and *E. muelleriana* used.

In the event of fire, adequate seed storage in the eucalypt canopy is a prerequisite for regeneration, but even so seed dispersal is likely to be limited to much less than 4–5 tree heights from the source (Ashton 1986). Small areas of scrub, therefore, may be expected to be encroached by eucalypts after fire but this will be a rare event in non-fire periods. Large areas of scrub, therefore, are likely to remain an intractable problem, even if fires of suitable frequency and type occur at times coinciding with abundant eucalypt seed production.

The management of these areas for the ultimate aim of restoration of forest would require the removal or control of the damaging herbivore population, which would be an anathema to the policy of a National Park. The alternative of spot sowing or planting appropriate eucalypt species with fencing guards would provide focal points from which subsequent regeneration may occur under the right conditions of disturbance. However, any measures would be successful only in the long-term because of the magnitude of the area of forest that has been devastated during a century of uncontrolled fire regimes.

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